

Internet of Things: Protocols and System Software

Los Del DGIIM, losdeldgiim.github.io

Doble Grado en Ingeniería Informática y Matemáticas
Universidad de Granada

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Granada, 2026

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1. Introduction

In the current technological world, there are three different main trends:

1. Computing: computers are getting smaller, cheaper and more powerful.
2. Sensing: there is a huge variety of sensors that can be used to measure different physical properties. They are also getting smaller and cheaper.
3. Transmission: two factors are being taken into account:
 - There are more and more wirelesscommunication technologies, with different characteristics. This leads to omnipresent connectivity.
 - Energy consumption is a key factor in the design of these technologies, as many IoT devices are battery-powered. This is leading to the development of low-power communication protocols and energy-efficient hardware.

In this context, the Internet of Things (IoT) is emerging as a paradigm where everyday objects become smart and connected to the internet.

Definición 1.1 (IoT Thing). An IoT thing is a physical object that is augmented with embedded electronics, making it therefore “smart”.

1.1. Digital Twin

The Digital Twin is a global representation of the world, where every physical object has a digital counterpart. This means that every physical object is represented in the digital world, and every change in the physical world is reflected in the digital world and vice versa. This is a very ambitious vision, and it is still far from being achieved. However, it is a useful concept to understand the potential of IoT and the challenges that it poses. Its three main features are:

- Live: the digital twin is continuously updated with data from the physical world.
- Predictive: the digital twin can be used to predict the behavior of the physical world using simulations and machine learning techniques.
- Interactive: depending on the predictions made by the digital twin, it can change and therefore make changes in the physical world.

In this situation, the world would become observable and programmable in real time, which would have a huge impact on many different fields, such as healthcare, transportation, manufacturing, etc. However, there are still many challenges that need to be overcome to achieve this vision.

1.2. IoT Views

There are two different views on IoT:

- IoT as a network of connected *things*, aka *M2M (Machine to Machine)*.

It is quite a low-level technical view, where the focus is on the devices and the communication between them. Internet is just seen as a communication medium, and the main goal is to connect as many things as possible.

It is the view that is usually taken by engineers and computer scientists.

- IoT as a network of connected *data*, aka *Big Data*.

It is a more high-level view, where there are still things but the focus is on the data that is generated by these things and how it can be used to create value. Internet is seen as a platform for data collection and analysis, and the main goal is to extract insights from the data.

It is the view that is usually taken by business people and data scientists.

2. IoT Systems

An IoT system consists of all the HW and SW components that are required to implement an IoT application. The components of an IoT system can be categorized into three main groups: devices, communication, and data processing. In order to reduce the complexity of an IoT system, developers tend to reuse the same components across different applications, and here is where the concept of *IoT platforms* comes into play.

2.1. IoT Platforms

An IoT platform is a framework that provides a set of tools and services to facilitate the development, deployment, and management of IoT applications. By using an IoT platform, developers can focus on the application logic rather than dealing with the complexities of the underlying infrastructure. There are many IoT platforms available in the market, each with its own strengths and weaknesses. There are two main categories of IoT platforms:

- Horizontal: These are general-purpose platforms that can be used for a wide range of IoT applications. They provide a broad set of features and services that can be customized to fit the specific needs of different applications.
- Vertical: These platforms are designed for specific industries or use cases. They provide specialized features and services that are tailored to the requirements of a particular domain.

2.2. IoT System Architecture

An IoT system architecture defines whole IoT system. Regarding the HW, it consists of 4 different types of components:

1. IoT Things: Also known as Smart Objects, these are the physical devices that actually interact with the environment.
2. Gateways: These are devices that act as intermediaries between the IoT Things and the Internet. Given that the IoT Things are often resource-constrained and may not have the capability to connect directly to the Internet, IoT Gateways provide the necessary connectivity and protocol translation.
3. Internet: This is the communication infrastructure that allows the IoT Gateways to connect and exchange data with other devices and services.

4. Online Services: These are cloud-based platforms and services that provide different functionalities to support the IoT applications.

However, SW components of an IoT system are also crucial to the overall functionality and performance of the system. There are three main aspects from the SW perspective that affect the whole IoT system.

2.2.1. SW Functional Blocks

From the SW perspective, it is essential to identify the functional blocks that are required to implement an IoT application. It is difficult to define a standard set of functional blocks that can be applied to all IoT applications, but for the aim of this course, we will consider three functional blocks:

1. IoT Networking: SW necessary to enable communication between all the components of the IoT system.
2. IoT Data Management: SW responsible for measuring, collecting, storing, and making data available for querying and analysis.
3. IoT Data Processing: SW responsible for analyzing the data collected from the IoT Things and extracting useful insights and information.

There are two main recipient groups of the data in order to make use of it:

- Humans: The data is presented to the users in a human-readable format, such as dashboards, reports, or visualizations.
- Algorithms: The data is used as input for machine learning models, predictive analytics, or other algorithms that can make decisions or take actions based on the data.

There are four main levels of data processing that can be applied to the data collected from the IoT Things:

- a) Descriptive: The data is only used to observe and describe the current state of the system.
- b) Diagnostic: The data is used to identify the causes of certain events or conditions in the system and explain why they happened.
- c) Predictive: The data is used to anticipate future events or conditions in the system based on historical data and patterns.
- d) Prescriptive: The data is used to perform actions or make decisions based on the insights derived from the data analysis. This has several legal and ethical implications, and usually requires human supervision and intervention to ensure that the actions taken are appropriate and that someone is legally responsible for the consequences of those actions.

2.2.2. SW Blocks Placement

The SW functional blocks can be executed in different specific HW components within the IoT system, depending on the specific requirements and constraints of the application. There are three main approaches for the placement of the SW functional blocks.

1. Cloud-based IoT Systems: The “classic” approach.

In this approach, the biggest part of the three SW functional blocks are executed in the cloud.

- IoT Things are responsible for collecting data and sending it to the gateway. Therefore, they only execute a small part of the IoT Networking and IoT Data Management blocks.
- IoT Gateways are responsible for receiving the data from the IoT Things and forwarding it to the cloud. Therefore, they execute a small part of the IoT Networking.
- The online services in the cloud are responsible for executing the majority of the three SW functional blocks, including IoT Networking, IoT Data Management, and IoT Data Processing.

2. Edge-based IoT Systems: The “modern” approach.

This approach is similar to the cloud-based approach, but with a significant shift of the SW functional blocks towards the gateways. Given that the online services have limited resources and are under high load, it is necessary to offload some of the processing tasks to the gateways in order to reduce latency and improve performance.

3. Mesh-based IoT Systems: The “future” approach.

In this approach, the SW functional blocks are even for shifted towards the IoT Things, which are becoming more powerful and capable of executing more complex tasks. This approach just considers gateways and online services as fallback options for the IoT Things in case they are not able to execute certain tasks.

2.2.3. SW Blocks Communication

The SW functional blocks need to communicate with each other in order to exchange data and coordinate their actions. This communication can be achieved through different connectivity options and protocols, depending on the specific requirements and constraints of the application. We will explore them in more detail in the rest of the course.

3. IoT Internetworking

During this chapter, we will focus on the IoT internetworking aspect, which is crucial for enabling communication and data exchange between different components of an IoT system. All the different components of an IoT system need to be able to communicate with each other in order to function properly, and the IoT Things need to be able to send the measured data and receive commands and updates from the users and applications. Therefore, connectivity and communication protocols are essential for the overall functionality and performance of an IoT system.

A reader may initially think that using the Internet and the well-known TCP/IP protocol suite would be sufficient for enabling communication in an IoT system, and it is in fact a good approach as these protocols have proven to be extremely successful and widely adopted in the traditional Internet. However, there are several challenges and limitations that need to be addressed when using the Internet and TCP/IP protocols in the context of IoT systems, as an usual Internet device is very different from an IoT device in terms of resources, capabilities, and requirements.

- Data Item Size: IoT devices often generate and exchange small data items, such as sensor readings or control commands, while the Internet and TCP/IP protocols are designed to handle larger data items, such as web pages or multimedia content.
- Not Optimized for Energy Efficiency: IoT devices are often battery-powered and have limited energy resources, while the Internet and TCP/IP protocols are not optimized for energy efficiency and may require more power consumption than what is feasible for IoT devices.
- Large Buffer Requirements: The Internet and TCP/IP protocols may require larger buffer sizes than what is available on IoT devices, which can lead to performance issues and increased latency.
- Pull vs. Push Communication: The Internet and TCP/IP protocols are primarily designed for pull-based communication, where clients request data from servers, while IoT systems often require push-based communication, where devices need to send data to other devices or applications without waiting for a request.

In order to address these challenges and limitations, there are several IoT-specific protocols and technologies that have been developed to enable efficient and effective communication in IoT systems. There are two main approaches to connect the IoT Things to the Internet:

- Application layer proxy:

There is a proxy (usually implemented in the IoT gateway) that translates between the protocols used by the IoT Things and the protocols used by the Internet.

- The IoT Thing communicates with the proxy using dedicated IoT protocols that have nothing to do with the Internet protocols.
- The proxy stores the data received from the IoT Things and makes it available to the Internet using standard Internet protocols.

An example of this approach is when an user has to install a specific app on their smartphone in order to interact with a smart home device, such as a smart thermostat or a smart light bulb.

- IP Router:

In this approach, the IoT Things directly use IP, but with some optimizations and adaptations to address the challenges and limitations mentioned above. An usual router may be combined (or may replace) with the IoT gateway to enable the IoT Things to connect to the Internet using IP. That way, the IoT Things can communicate directly with other devices and applications on the Internet without the need for a proxy, which is much more scalable. There is no need for specific apps or software to interact with the IoT Things, as they can be accessed using standard Internet protocols and interfaces.

During the rest of the course, we will explore in more detail the second approach, which is the one that is becoming more and more popular in the IoT ecosystem, as it provides better scalability and interoperability.

3.1. IEEE 802.15.4

IEEE 802.15.4 is a standard for low-rate wireless personal area networks (LR-WPANs) that is widely used in IoT applications. Its main focus is on providing low-power, low-cost and low-complexity wireless communication for IoT devices. It defines the Physical Layer (PHY) and part of the Data Link Layer (MAC). In addition, it defines multiple variants for each of the layers, which can be used in different scenarios and applications depending on the specific requirements and constraints of the IoT system.

Communication using this standard is based on a IEEE 802.15.4 PAN, which is a collection of devices that are connected together using the IEEE 802.15.4 standard. The members of a PAN can either be:

- Full Function Devices (FFDs): These devices can communicate with any other device in the PAN, including other FFDs and Reduced Function Devices (RFDs). They can also take over network management activities.

If two devices need to communicate with each other but they are not in range of each other, they can use an FFD to forward messages between them, which

allows for multi-hop communication and extends the range of the PAN. Only FFD have this capability.

- Reduced Function Devices (RFDs): These resource-constrained devices can only communicate with one FFD in the PAN, and they cannot take over network management activities.

In addition, each of the members of a PAN can have different roles, which determine their behavior and responsibilities in the network:

- PAN Coordinator: This is the device that is responsible for managing the PAN, including tasks such as the PAN creation (by selecting a PAN ID and a channel), adding and removing devices from the PAN, and coordinating the communication between devices in the PAN. There can only be one PAN Coordinator in a PAN, and it must be an FFD.
- Coordinator: This is a device that is responsible for managing a portion of the PAN, such as a cluster of devices. It can also take over some of the network management activities, such as routing and forwarding of messages. There can be multiple Coordinators in a PAN, and they must be FFDs.
- Members: These are the devices that are part of the PAN and can communicate with other devices in the PAN. They can be either FFDs or RFDs, and they do not have any specific responsibilities or roles in the network.

Once the members of a PAN have been defined and assigned their roles, they can communicate with each other using one of these two types of topologies:

- Star Topology: In this topology, all the devices in the PAN communicate directly with the PAN Coordinator, which acts as a central hub for the communication. This topology is simple and easy to manage, but it can be less efficient and less scalable than other topologies, as all the communication has to go through the PAN Coordinator, which can become a bottleneck and a single point of failure.
- Peer-to-Peer Topology: In this topology, the devices in the PAN can communicate directly with each other without the need for a central hub. This topology is more efficient and scalable than the star topology, as it allows for direct communication between devices, which can reduce latency and improve performance. However, it can be more complex to manage, as it requires more coordination and synchronization between devices to ensure that messages are delivered correctly and efficiently.

It should however be noted that the IEEE 802.15.4 standard does not define a protocol but only a standard for the physical and data link layers, so specific protocols as Zigbee have been developed on top of it to provide all the necessary functionalities and features for enabling communication in IoT systems.

3.1.1. Physical Layer (PHY)

The Physical Layer (PHY) of the IEEE 802.15.4 standard defines the physical characteristics and parameters of the wireless communication, such as the frequency bands where the communication can take place. It defines three different frequency bands that can be used for communication:

- 868 MHz Band: This band is used in Europe and other regions, and it provides a data rate of up to 20 kbps. It has one channel available for communication.
- 915 MHz Band: This band is used in North America and other regions, and it provides a data rate of up to 40 kbps. It has ten channels available for communication.
- 2.4 GHz Band: This band is used worldwide, and it provides a data rate of up to 250 kbps. It has sixteen channels available for communication.

The services provided by the PHY layer are divided into two categories:

- Data Services: These services are used for transmitting and receiving data between devices in the PAN.
- Management Services: These services are used for managing the communication and the network, such as channel scanning (to find the best channel for communication), energy detection (to measure the energy level of the channel and determine if it is busy or idle), and link quality indication (to measure the quality of the communication link between devices).

The content that is in fact transmitted over the air is the *PHY Protocol Data Unit (PPDU)*, which consists of the *PHY Service Data Unit (PSDU)* encapsulated with the header. The 6 bytes Header consists of:

- Synchronization Header (SHR), 5 bytes:
This header is used for synchronization between the transmitter and the receiver, and it consists of a preamble (4 bytes) and a start of frame delimiter (1 byte).
- PHY Header (PHR), 1 byte:
The first 7 bits of this header are used to indicate the length of the PSDU, while the last bit is reserved for future use. Therefore, the maximum length of the PSDU is $2^7 - 1 = 127$ bytes.

3.1.2. Data Link Layer (MAC)

3.2. NB-IoT

3.3. 6LoWPAN

3.4. MQTT

4. Question Sheets

4.1. Introduction

The recorded lecture is provided [here](#).

Ejercicio 4.1.1. What are the two views on IoT discussed in the lecture? How do they differ?

The two views on IoT are:

- IoT as a network of connected *things*, aka *M2M (Machine to Machine)*.
- IoT as a network of connected *data*, aka *Big Data*.

They differ in plenty of different aspects:

- While the first one focuses on the devices and the communication between them, the second one focuses on the data that is generated by these things and how it can be used to create value.
- While the first one sees Internet as just a communication medium, the second one sees Internet as a platform for data collection and analysis.
- While the main goal of the first one is to connect as many things as possible, the main goal of the second one is to extract insights from the data.
- While the first one is usually taken by engineers and computer scientists, the second one is usually taken by business people and data scientists.

Ejercicio 4.1.2. Are IoT gateways always necessary? What are they used for?

IoT gateways are used to connect IoT things to the internet. IoT things are usually resource-constrained devices that cannot connect directly to the internet, so they need a gateway to act as a bridge between them and the internet. However, IoT gateways are not always necessary, as some IoT things can connect directly to the internet using low-power communication technologies such as Wi-Fi or cellular. In this case, the IoT thing itself can act as a gateway, and there is no need for an additional device.

4.2. IoT Systems

The recorded lecture is provided [here](#).

Ejercicio 4.2.1. What is the difference between an IoT system and an IoT platform? Use the WWW to find another IoT platform that is not mentioned in the lecture.

They are different concepts. An IoT system consists of all the HW and SW components that are required to implement an IoT application, and developers tend to reuse the same components across different applications. This is where the concept of IoT platforms comes into play, because an IoT platform is a framework that provides a set of tools and services to facilitate the development, deployment, and management of IoT applications. By using an IoT platform, developers can focus on the application logic rather than dealing with the complexities of the underlying infrastructure. Therefore, an IoT platform is a tool that can be used to build an IoT system.

An example of an IoT platform that is not mentioned in the lecture is [Blynk](#). It is an horizontal IoT platform that provides a set of tools and services to facilitate the development, deployment, and management of IoT applications. It allows developers to create custom IoT applications using a drag-and-drop interface, and it supports a wide range of hardware platforms and communication protocols.

Ejercicio 4.2.2. Is the Microsoft Azure IoT Suite a horizontal or a vertical IoT platform? Why?

It is a horizontal IoT platform because it provides a broad set of features and services that can be customized to fit the specific needs of different applications. It is not designed for a specific industry or use case, but rather it can be used for a wide range of IoT applications across different domains.

Ejercicio 4.2.3. Provide an example (i.e., search on the Internet or come up with your own idea) for a scenario in which you would use predictive data processing but not prescriptive data processing. What is the main legal difference between these?

There are plenty of cases where predictive data processing can be used without prescriptive data processing for several reasons:

- Prescriptive data processing can simply not be made.

For instance, an IoT application designed to predict the weather conditions in a specific area based on historical data and patterns. The application can provide valuable insights and information about the expected weather conditions, but the weather cannot be changed by the application, so prescriptive data processing is not possible in this case.

- Prescriptive data processing can be made, but it is not ethical to do it without human supervision and intervention.

For instance, an IoT application designed to predict the likelihood of a patient developing a certain medical condition based on their health data and lifestyle habits. The application may recommend certain lifestyle changes or medical interventions to reduce the risk of developing the condition, but it is not ethical to implement these recommendations without human supervision.

- Prescriptive data processing can be made, but a responsible person must be legally accountable for the consequences of the actions taken based on the prescriptive data processing.

For instance, an IoT application designed to predict the future of the stock market based on historical data and patterns. The application may recommend certain investment strategies to maximize returns, but someone should be legally accountable for the consequences of the actions taken based on these recommendations, as the stock market is highly volatile and unpredictable, and there is a risk of financial loss.

The main legal difference between predictive and prescriptive data processing is that prescriptive data processing actually performs actions or makes decisions, and they can have legal consequences, so there must be a responsible person who is legally accountable for those consequences.

Ejercicio 4.2.4. What are the differences between mesh-based and edge-based IoT systems? Which one is currently supported by platform providers?

Both IoT system architectures are designed to process data closer to where it is generated, but in the edge-based architecture the IoT things are still really simple and resource-constrained, so they just collect data and send it to the gateway, which is responsible for processing the data and making decisions based on it. In contrast, in the mesh-based architecture, the IoT things are more powerful and can communicate directly with each other, so they can process data and make decisions without the need for a central gateway.

Currently, even though the most supported architecture by platform providers is the cloud-based one, there are some platforms that support edge-based IoT systems, but mesh-based IoT systems are still not widely supported by platform providers due to the complexity and challenges associated with implementing and managing such systems.

4.3. IoT Internetworking

4.3.1. Motivation

The recorded lecture is provided [here](#).

Ejercicio 4.3.1. Why do we need new protocols for the IoT instead of using established Internet protocols? Describe one challenge existing protocols face.

New protocols are needed for the IoT because existing Internet protocols, such as TCP/IP, are not optimized for the resource-constrained nature of IoT devices. One challenge existing protocols face is that they may require larger buffer sizes than what is available on IoT devices, which can lead to performance issues and increased latency.

Ejercicio 4.3.2. Which central protocol can we not replace (completely) for the IoT? Why is it harder to replace it than others?

The central protocol that we cannot replace (completely) for the IoT is the Internet Protocol (IP). It is harder to replace IP than other protocols because it is the fundamental protocol that enables communication and data exchange between different devices and applications on the Internet. It is the backbone of the Internet and, technically speaking, being connected to the Internet requires having an IP address and using IP for communication. Therefore, while we can use different protocols for the application layer or the transport layer, we still need to use IP at the network layer in order to enable connectivity and communication with the Internet.

Ejercicio 4.3.3. We introduced two approaches for Smart Things to connect to the Internet. Name these approaches, describe them briefly and give one advantage and one disadvantage of each, compared to the other.

The two approaches for Smart Things to connect to the Internet are:

- Application layer proxy: In this approach, there is a proxy (usually implemented in the IoT gateway) that translates between the protocols used by the IoT Things and the protocols used by the Internet. The IoT Thing communicates with the proxy using dedicated IoT protocols that have nothing to do with the Internet protocols, and the proxy stores the data received from the IoT Things and makes it available to the Internet using standard Internet protocols.
 - Advantage: It allows for the use of dedicated IoT protocols that are optimized for the resource-constrained nature of IoT devices, which can lead to better performance and energy efficiency.
 - Disadvantage: It introduces an additional layer of complexity and can lead to increased latency and reduced scalability (due to the dedicated software), as all communication between the IoT Things and the Internet needs to go through the proxy.
- IP Router: In this approach, the IoT Things directly use IP, but with some optimizations and adaptations to address the challenges and limitations mentioned above. There is no need for specific apps or software to interact with

the IoT Things, as they can be accessed using standard Internet protocols and interfaces.

- Advantage: It provides better scalability and interoperability, as the IoT Things can communicate directly with other devices and applications on the Internet without the need for a proxy.
- Disadvantage: It may require more complex and resource-intensive protocols and technologies to enable efficient communication and data exchange between the IoT Things and the Internet, which can lead to increased power consumption and reduced performance for resource-constrained IoT devices.

4.3.2. IEEE 802.15.4

The recorded lectures are provided [here \(Part A\)](#) and [here \(Part B\)](#).

Ejercicio 4.3.4. What are the differences between the PHY and the MAC layer?

Both are layers in the OSI model, but they serve different purposes. The PHY (Physical) layer is responsible for the physical transmission of data over the communication medium. The MAC (Medium Access Control) layer builds on top of the PHY layer and is responsible for managing access to the communication medium, including addressing, framing, and error detection.

Ejercicio 4.3.5. Discuss pros and cons of using 16 bit addresses for 802.15.4 instead of 64 bit addresses.

Ejercicio 4.3.6. Describe the basic idea of CSMA/CA and compare it to frequency hopping. You may use a diagram / drawing to describe your answer.

Ejercicio 4.3.7. Briefly describe and compare TDMA and FDMA. Which one is better?

Ejercicio 4.3.8. An 802.15.4 network is deployed in the following three application scenarios. Which CSMA/CA MAC Layer version would you use? Justify your answer (e.g., “In this case, we should use slotted CSMA/CA, because...”).

- A smart home network, where only 10 temperature sensors are installed. Each sensor reports a measurement once per minute.
- A smart clinic network that transmits monitoring data about the patients and alarm notifications.

Ejercicio 4.3.9. When unslotted CSMA/CA is applied, at each trail the back-off window size will be doubled. However, it doesn't always lead to a longer delay, why?